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Intelligent safety control modeling and construction process control of steel mesh frame structures

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ABSTRACT

In the process of large-span steel structure construction, because its structure has not yet formed a complete system, so its mechanical properties and safety status are not stable, and it is necessary to control the safety of the structure, and the current traditional structural health monitoring (SHM) has the problems of high cost, easy to be interfered, and unpredictable, etc. The digital twin, as a popular technology in the engineering industry, has the characteristics of multidimensionality, high integrative, intelligent and real-time, so it is very necessary to introduce digital twin technology in the construction process. In this paper, a digital twin-driven intelligent safety control method for large-span steel structures is proposed. The digital twin model is constructed by collecting actual engineering data and integrating finite element analysis data. Based on the random forest algorithm, The influencing factors of the overall jacking construction process of the large-span steel mesh frame are analyzed to obtain the importance ranking of the features affecting the process. The results show that the most influential features for the overall jacking construction are the height difference of the jacking frame and the jacking speed. Therefore, the above two should be paid special attention to in the construction safety control.

1. Introductions

Space steel mesh frame structure is one of the earliest and most widely used structural forms of various space structures, which is commonly used in the roofing system of some large-scale infrastructures and public buildings, and is often seen in some airports [1,2], high-speed rail stations [3], gymnasiums [4], and exhibition centers and other architectural facilities. The mesh frame structure itself has excellent structural properties such as good integrity, high spatial stiffness, strong stability, light weight and beautiful shape [5]. However, since the full stress design method is generally used for calculation and analysis in the design stage of the net frame structure, this also leads to the fact that the net frame structure is very sensitive to the initial defects of the rods and the adjustment of the use of the load and other factors, and in the process of construction or operation and maintenance of the construction load [6], the use of the function of the changes in the use of the renovation of the event, resulting in the load borne is different from the design stage, so that the key rods and the key node The load carrying capacity is exceeded, leading to local instability or overall collapse of the structure, which ultimately causes major casualties and economic losses, therefore, the control of structural safety is particularly important.

Currently, general construction projects collect the safety data of the structure through contact sensors, total stations [7], etc., and compare the data with the limit values corresponding to the code, so as to derive the current safety status of the structure, i.e., the structural health detection (SHM) in the traditional sense. However, this method has certain drawbacks: (1) structural health detection requires the installation and maintenance of a large number of sensors, but due to the economic impact, they are often installed at critical locations only based on engineering experience, and both periodic inspection and data analysis require a large amount of manpower and material input, making the cost of the traditional SHM method relatively high [8,9]; (2) the contact sensors usually need to be manually mounted directly on the structure. Some of the sensors have inconvenient installation problems [10], which may lead to inaccurate data collection due to irregular installation factors, and they may also be dropped outdoors due to wind, rain, snow, and human influence; (3) The traditional SHM method mainly focuses on the current state of the structure [11], which makes it difficult to accurately predict the future performance of the structure and the possible problems that may occur. However, Wang et al. have proposed some effective methods for engineering nonprobabilistic transient reliability in the case of insufficient data[12–15].

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Digital twin technology is to make full use of physical models, sensor updates, historical operational data, etc., integrating multi-disciplinary, multi-physical quantity, multi-scale, multi-probability simulation processes, and completing the mapping in the virtual space, so as to reflect the corresponding entity's full life cycle process. Pioneered in the aerospace field and applied, in recent years, digital twins have also been introduced in the manufacturing industry [16,17], construction industry [18,19] to carry out digital intelligence upgrading, especially in the introduction of digital twins in civil engineering projects, the digital twins are used in the whole life cycle of the project, such as design [20], construction [21,22], operation and maintenance [23,24], demolition remodeling [25], and other phases. Especially in the construction phase of civil engineering projects, digital twin technology has been more widely used.

In the context of bridges and highways, Mafipour [26] proposed a reverse engineering method for parametric modeling of bridges. for generating parametric models of bridges from segmented point clouds. Following the concept of reverse engineering for parametric modeling, a parametric prototype model (PPM) is proposed as a tool for extracting parameter values from the point cloud. Local and global optimization problems were defined to tune the PPM and assemble it into an integrated model. Jiang et al. [27] proposed a digital twin based road widening system approach for checking the clearance of underpasses in highway widening projects using online map data, creating digital twins of underpasses based on the digital twins of existing arterials and a BIM model of the newly widened road to aid in clearance inspection and redesign.

In terms of tunneling, Li et al. [28] conducted a comprehensive analysis from the perspectives of real-time or instantaneous data acquisition, data interpretation, and data aggregation in a complete and seamless "data acquisition-interpretation-aggregation" workflow, which is linked to dynamic tunneling to support optimal design, and proposed a conceptual framework for instantaneous tunneling support. Design. Subsequently, it integrates digital twins (DT) into the implementation of JIT design for rock tunnels, proposes a scenario-entity-relationship-event-control modeling approach as a novel DT application model for cross-domain sharing and reuse, and integrates multiple advanced data acquisition, interpretation, and aggregation methods using DT [29].

In terms of building structure, Liu et al. [30] proposed an intelligent construction method based on digital twin, which integrates IoT, BIM and finite element model to build a framework of intelligent construction method based on digital twin, gives real-time feedback and regulation to the whole process of construction, and improves the information utilisation rate of construction operation.Wang et al., based on the concept of digital twin, proposed a method to consider the spatio-temporal variation of the Based on the concept of digital twin, Wang et al. proposed a safety control method for cable network structure considering spatial and temporal changes. This method can monitor and control the construction process of cable network in real time on site, and it is convenient to compare and analyse the mechanical and geometrical information on the construction site with the real-time finite element simulation results [31].

In order to solve the technical problems of structural health monitoring and digital twin, this study combines the digital twin (DT) technique with the stochastic Morien algorithm. It makes up for the traditional SHM method that relies heavily on manual operation and judgement, which leads to errors and subjectivity, and solves the difficult real-time problem, and also makes the digital twin technology complete the mirror mapping of the structure with the help of sensors.

Based on this, this paper proposes a digital twin framework for safety control of spatial structures based on random forests, and verifies the effectiveness of the proposed method by establishing a model of random forests, using actual construction data for training and feature correlation analysis. This study aims to achieve the purpose of risk control based on digital twin and machine learning technology, with the help of Random Forest algorithm to process the data collected during the construction of the overall jacking of the steel mesh frame and to improve the next construction step. The method has the following advantages: (1) low cost - through the reasonable arrangement of sensors, it can achieve the monitoring of the key position of the structure and establish the digital twin model through the data; (2) real-time - combined with the historical data, it can be realised in DT space health monitoring, performance prediction and maintenance guidance; (3) high accuracy - able to dynamically show the mapping of the physical structure in DT space through the sensors; (4) expandability - able to be carried out in engineering projects using the integral jacking method and the integral jacking method for use.

The rest of the paper is organised as follows: Section 2 describes the method of building an intelligent safety control framework based on digital twin technology for large-span steel mesh frame structures. Section 3 describes the process and methodology of feature analysis based on the Random Forest algorithm, and Section 4 outlines the objectives and ways in which structural safety should be adopted for control after feature analysis, and proposes the simultaneous jacking control process and control system. Finally, Section 6 summarises the conclusions and potential applications.

2. Intelligent safety control method for large-span steel structures driven by digital twins

Digital twin technology refers to the simulation, analysis and optimisation of the whole life cycle of physical entities through digital means in order to achieve two-way mapping and interaction between physical entities and digital models. In the field of large-span steel structures, digital twin technology can realise real-time monitoring, data analysis, fault diagnosis and predictive maintenance of large-span steel structures, and provide powerful support for the safe and intelligent control of large-span steel structures [32].

In this paper, the DT framework for large-span steel structures is built, as shown in Fig. 1, which takes the space steel mesh frame structure as the physical entity for DT mapping, corresponding to the construction data of the project on the left side of the figure, including static data and dynamic data. Static data refers to the geometrical information of the relevant components of the structure, such as rod length, inner and outer diameters and monitoring equipment models, which are basically unchanged and inherent; dynamic data refers to the data that will change as the construction advances, such as ambient wind speed, temperature, and the data of rod stresses and displacements. It also contains a document database with architectural drawings, specifications and geographic information data, such as the top and bottom chord layout of the grid, structural construction staircase sample drawings and the underground pipe network of the terminal building. The right part is the digital twin model, which is a virtual object mapped by the physical space, containing geometric and finite element models (geometric, physical, behavioural, and rule-based), predictive control algorithms, and the physical building, which integrates all kinds of static and dynamic data.

2.1. Geometric models

To realise the perfect mapping of virtual space to physical space, it is necessary to construct a geometric model of the long-span steel structure first. The model should be able to comprehensively reflect the geometric characteristics and material properties of the long-span steel structure.

It mainly includes the construction form, span, height, material, connection form of nodes, etc. The material characteristics mainly include material type, yield strength, elastic modulus, etc., and it can achieve real-time synchronisation with the physical entity. In the construction process, the technical means of 3D scanning and laser measurement are used to obtain the accurate geometric information of the large-span steel structure. In this paper, three-dimensional modelling

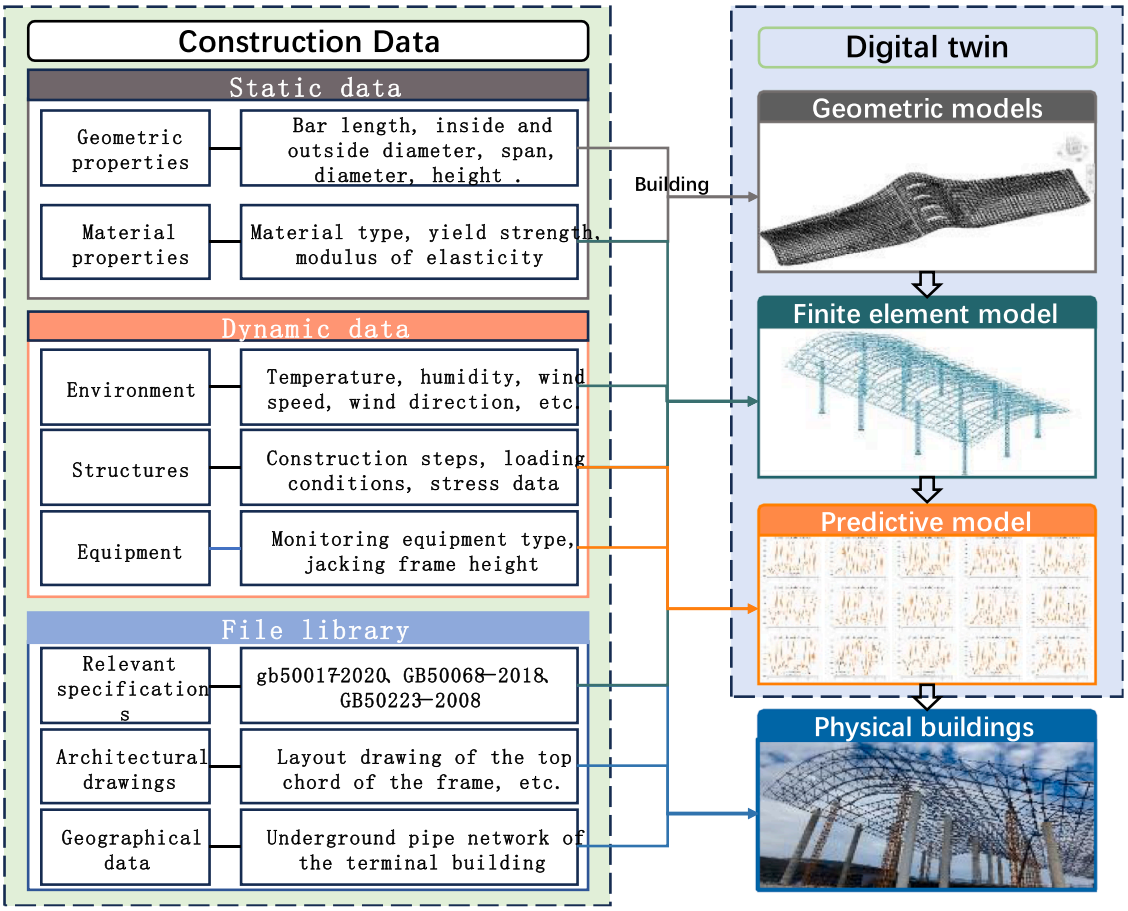


Fig. 1. DT frames for large-span steel structures.

is carried out through Tekla, Dynamo, Revit and other software to deepen the nodes and connection forms, and visual simulation is carried out on the spatial three-dimensional arrangement of the steel structure members to form a geometric model, as shown in Fig. 2.

2.2. Physical models

A physical model is a digital representation of a real-world physical entity or system that simulates the structure, properties, behaviour of physical entities and their interactions with the environment. The physical model is one of the core components of digital twin technology, which provides the basis for real-time synchronisation and interaction between the virtual space and the physical space.

In this paper, numerical simulation modelling has been carried out using finite element software, in which all components of the steel roof cover are meshed using beam cells. The model was designed completely based on the actual situation and boundary conditions. According to the characteristics of welded ball nodes and bolted ball nodes, a form of multiplying the self-weight coefficient of the steel roof cover by an expansion factor of 1.3 is used for application. The key to simulate the mechanical behaviour of the jacked structure is the actual estimation of the partial coefficient of static load, and a value of 1.3 was adopted for the finite element model. Fig. 3 shows the finite element model of the steel mesh roof during the jacking stage.

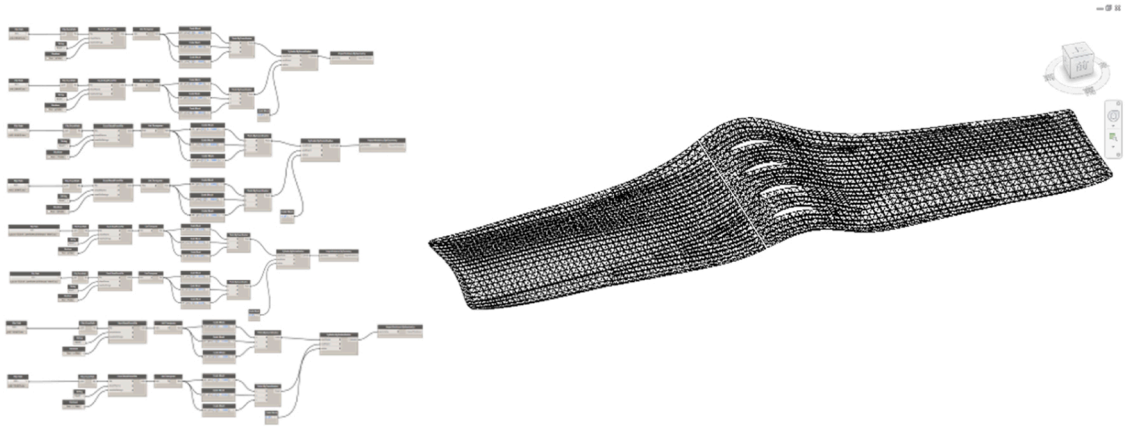


Fig. 2. Geometric models.

2.3. Behavioral models

At the behavioural level, digital twins are able to simulate the entire process of project construction, equipment operation or user behaviour. By building an accurate behavioural model, the mechanical response of a structure under different scenarios can be predicted and simulated, potential factors that may lead to risks can be identified, preventive measures can be taken in advance and solutions can be optimised.

In this paper, based on the geometric and physical models, the construction conditions consistent with the jacking scheme of the project are established in the behavioural layer, and the corresponding conditions are analysed and computed by the finite element model, which results in the loads that the project will be subjected to and the safety performance and structural state of the project under the construction conditions, and is used to predict the maintenance and safety control of the structure.

2.4. Rule-based model

In the whole digital twin modelling process, the rule model plays a crucial role. The rule model firstly defines and quantitatively limits the mechanical performance parameters of each type of equipment and size of components during the construction process according to the national, industrial or corporate internal standards. These parameters are the basis for assessing the safety performance of the structure, and any simulation results outside the limits will be considered unacceptable, thus ensuring the rigour of the analysis.

The rule-based model should also support the dynamic adjustment of mechanical performance parameters as construction progresses and external environmental conditions change. For example, considering the influence of temperature, humidity and other environmental factors on the material properties, the relevant parameters are automatically adjusted through the preset algorithms to make the simulation closer to the actual situation. Combined with the data monitored by the sensors in real time, the rule model is able to analyse the operating status of the equipment, predict the potential risk of failure, and send out warning signals in advance. This helps to take timely measures to avoid the adverse effects of equipment failure on construction progress and quality.

3. Data analysis and troubleshooting

Through the data actually collected in the project, fusing the twin data simulated by the high-fidelity digital twin model, the correlation analysis of the relevant parameters is carried out by taking the relevant contents of jacking height as the characteristic variables. This project is based on the random forest approach to carry out feature correlation analysis of the specific height of the jacking frame, the maximum jacking height, the average jacking height, the maximum height difference of the jacking, and the jacking speed, which provides certain basis for the safety control of the structure and the construction plan.

3.1. Random forest

The Random Forest (RF) algorithm is an integrated learning method that belongs to the category of supervised learning algorithms and consists of classifiers or regressors constructed from multiple decision trees. It was proposed by Leo Breiman [33] in 2001 and aims to improve the predictive accuracy and robustness of models by constructing multiple decision trees.

The Random Forest algorithm has demonstrated significant benefits and advantages in the field of machine learning and data analysis, and it stands out in many applications, mainly including: (1) By integrating multiple decision trees, Random Forest effectively improves the accuracy and stability of prediction, and this integration mechanism can significantly reduce the bias and variance of a single model, so as to obtain more reliable prediction results. (2) It performs well in overfitting resistance, thanks to its Bagging-based integration strategy and random feature selection mechanism, which helps to maintain the generalisation ability of the model in complex datasets. (3) Performs well in dealing with large-scale datasets, and its parallelised processing gives the algorithm a significant advantage in terms of efficiency and scalability.

In random forests, each decision tree is constructed by recursively partitioning the dataset into purer subsets. This partitioning is done based on a certain feature, with the aim of making the partitioned subset as low as possible in terms of impurity in the target variable (e.g., Gini impurity or information gain in classification problems, variance in regression problems). Therefore, each feature brings a certain amount of impurity reduction when splitting the nodes of the decision tree. The basic idea of average impurity reduction can be expressed by Eq.1:

$$MDI(X_m) = \frac{1}{N_T} \sum_{T \in Forest} \sum_{q \in T} p(q) \times \Delta i(s_q, q) \tag{1}$$

Where, X_m denotes a certain feature, N_T denotes the total number of decision trees in a random forest, T denotes a decision tree in the forest,

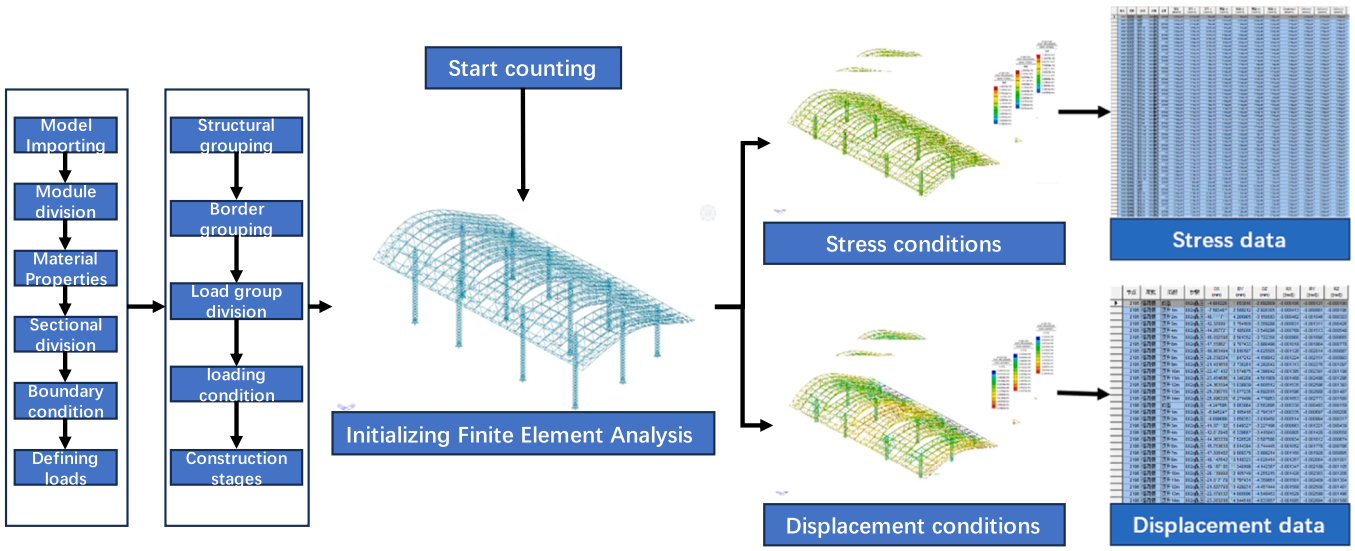


Fig. 3. Finite element modeling of steel mesh frame.

q denotes a node in the tree, $p(q)$ denotes the proportion of samples that are partitioned to node q among all the samples, s_q denotes the feature that is used in the splitting of node q, and $\Delta i(s_q, q)$ denotes the amount of change in the impurity before and after node q splits.

3.2. Information collection

In the construction process using the overall jacking method, the structure has not yet formed a reliable system, therefore, after consulting the corresponding steel structure construction specification and analysing the relevant factors affecting the safety of the structure, the stress is selected as the index for determining the performance of the structure, and the main influencing factors are the temperature effect, the wind load and the height of the jacking frame.

The overall jacking time is relatively short, in the process of the overall jacking temperature will not occur during this time, will not have a greater impact on the overall rigidity and stability of the structure; for the wind load, the jacking frame structure erection is completed, the top of the cable wind is also installed, as shown in Fig. 4 forming a more complete structural system, then start jacking, gravity load for the structure of the weight and the weight of the object to be lifted. In the construction specification, the structure should be able to resist when the wind is not more than grade 8, so it is not considered in this paper.

As the technology at the present stage cannot achieve a fully synchronised overall jacking, nor can it control the loading of cylinders and the height of a single jacking when jacking, the height of the jacking frame will have a slight error, which also leads to the part of the structure that does not match with the design deflection, and thus leads to changes in the structural stresses.

In summary, this paper chooses the height of the jacking frame as the main control content, i.e., the specific height of the jacking frame, the average height, the difference between the maximum height and the minimum height, and the jacking speed in the jacking process are the main influencing factors of the stress.

The specific height of the jacking frame refers to the height value of each jacking frame involved in jacking, and the overall jacking method is mainly to adjust the height of each jacking frame to realise the overall jacking, so the control of the specific height of the jacking frame is in fact the most important content to control the safety of the overall jacking method. In the specific construction process, the specific stiffness of the jacking frame is often not an intuitive factor to react to the structural safety performance, therefore, after analysing the specific height of the jacking frame, the average height of the jacking frames, the height difference between the highest jacking frame and the lowest jacking frame, and the speed of the jacking frame are more intuitive factors to react to the structural stresses.

The average height reflects the approximate height to which the roof was lifted by the jacking frames, i.e., the average position reached by all the jacking frames in the process of jacking, which is an important basis for evaluating the effect of jacking and determining the final position of the structure.

The height difference between the highest jacking frame and the lowest jacking frame reflects the height difference between different jacking frames. The size of the height difference has a direct impact on the stability and safety of the structure, so it needs to be closely monitored and adjusted during the jacking process.

The jacking speed reflects the height of each jacking step, which is an important index for assessing the jacking efficiency and controlling the stability of the jacking process. Too fast a jacking speed may lead to uneven stresses and reduced stability of the structure; while too slow a jacking speed may prolong the construction period and increase the construction cost. Therefore, it is necessary to control the jacking speed reasonably according to the actual situation during the jacking process.

Based on the importance of the jacked structure and the uncertainty of the overall jacking operation, during the jacking process, it is necessary to reflect the safety performance of the structure and the safety status of the structure through continuous monitoring indicators, mainly deflection, stress, strain, cracks and other mechanical property indicators. Since the project is a new construction stage and mainly a steel structure net frame, the crack factor of the rods can be disregarded, while the displacement and deflection of the adjacent points are reacted by the height of the jacking frame, which is a factor affecting the stress, and is no longer considered as a separate factor. Therefore, during the overall jacking process, the strain of the key bars is monitored and the bar stresses are calculated to assess the reliability of the structure and the safety performance of the construction.

The sensor network can monitor the large-span steel structure in real time and collect various data including stress, strain and temperature. The collected data can also be pre-processed, including filtering, denoising, calibration and other operations to improve the accuracy and reliability of the data. In this project, stresses are measured and monitored by deploying vibrating string stress sensors on the critical bars of the steel mesh frame to react to the safety performance of the structure, as shown in Fig. 5.

Stress data were collected and stored automatically by a computer-controlled data acquisition system. Careful consideration was given to the selection of monitoring points, and based on the analysis of the simultaneous jacking simulation, vibrating string stress gauges were mounted on the surfaces of key bars, making it relatively easy to install monitoring points as well, ultimately in order to collect sufficient stress values representative of the overall steel roof with a suitable number of sensors.



Fig. 5. Stress sensors on rods.

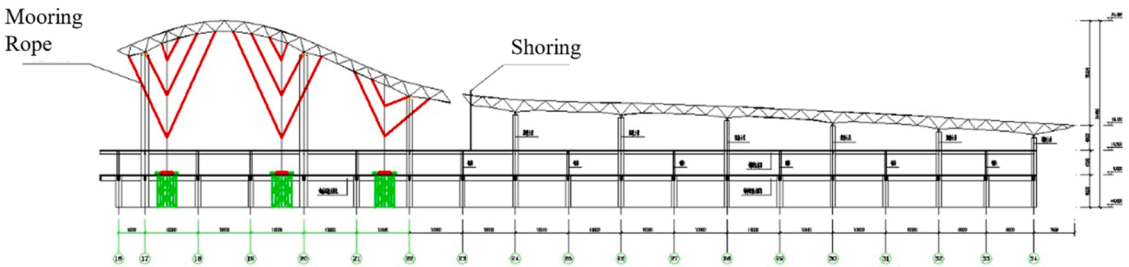


Fig. 4. Equipment Layout Schematic.

3.3. Data preprocessing

3.3.1. Missing value processing

In the data preprocessing stage, missing value processing is one of the first tasks. In this paper, the linear interpolation method is adopted as the main missing value interpolation strategy, especially for variables with low missing proportions. Linear interpolation effectively fills the data gaps by using neighbouring non-missing observations for estimation, while maintaining the continuity and trend of the data [34]. The linear interpolation formula used in this paper is shown in Eq. 2

$$y = y_0 + k(x - x_0), \quad k = \frac{y_1 - y_0}{x_1 - x_0} \tag{2}$$

3.3.2. Outlier processing

Outliers, as extreme observations in the data, may have a significant impact on the results of the analyses. Therefore, various statistical methods (Z-score test, IQR method) were used to identify outliers in the dataset during the data cleaning process. For the detected outliers, appropriate treatment strategies were adopted based on the specifics of the data and analysis needs. Obvious errors or outliers that could not be reasonably explained without relevant data before and after were deleted; outliers that were caused by similar data before and after, by measurement errors or data entry errors were corrected by linear interpolation through the proximity of the data; and outliers that deviated from the subject distribution but had potential analytical value were retained and analysed as special cases for subsequent analyses.

3.3.3. Denoise

In order to further improve the data quality and to reduce random noise and errors in the data, data noise reduction was performed on specific variables by sliding mean filtering method. Sliding mean filtering, as a simple but effective low-pass filter, achieves the suppression of high-frequency noise components and the retention of low-frequency signal components by weighted averaging of observations and their neighbouring observations. During the filtering process, a sliding window m= 10 is set to ensure that the noise is removed without losing important information of the signal. The data noise reduction formula used in this paper is shown in Eq. 3.

$$y_i = \frac{1}{m} \sum_{k-m/2}^{m/2} x_{i+k}, \quad m = 10 \tag{3}$$

3.3.4. Characteristic Importance Analysis

In this paper, the following steps are taken to carry out the analysis of feature correlation and feature importance based on random forest:

Constructing Decision Trees: Firstly, each tree in a random forest is constructed independently. When constructing each tree, the random forest algorithm randomly selects a subset of features, and searches for the optimal split points on these features to construct the nodes of the tree.

Node Gain: Whenever a feature is used to split a node, the algorithm calculates the gain generated by the split, i.e., the amount of impurity reduction before and after the split using the feature. Impurity is usually measured by Mean Squared Error (MSE).

Feature Contribution Accumulation:For each tree, the algorithm keeps track of the contribution of each feature in the split nodes and accumulates these contributions to get the total contribution of the feature.

Average Feature Importance: Since the random forest contains multiple trees, the contribution of each feature is averaged across all the trees to get the average importance score of the feature.

Feature Importance Ranking: Finally, all features are ranked in order of importance to get the highest to lowest ranked feature importance and to get the most important features that affect the prediction results.

3.3.5. Characteristic correlation

Fig. 6 shows the correlation matrix between the input feature variables considered in this study, and it can be observed that there is a strong correlation between each specific jacking height (R=1),which is mainly attributed to the fact that the jacking heights are actually interacting with each other, because the specification requires that the difference between the jacking heights is required to be within 10 mm, and the specific heights of the individual jacking frames during the jacking process are also unevenly affected because the maximum height of that jacking is not reached occasionally, so the maximum height difference and the jacking speed are also affected by each jacking height. Occasionally reached the highest height of the jacking, so the maximum jacking height and the average jacking height is a uniform effect, and the maximum height difference and jacking speed is also affected by the individual jacking height.

3.3.6. Characteristic importance

Determining the order of importance of each feature is critical to the impact on the structural safety performance of the overall jacking, and the results of the feature importance analysis can provide a basis for the safety control of the overall jacking phase. The Random Forest algorithm is used to assess the feature importance by disrupting the order of the features and calculating the degree of attenuation of the comparative model performance. The trained Random Forest model is analysed several times according to the sensors at different locations to obtain the importance ranking of the features, as shown in Fig. 7.

From Fig. 7, it can also be concluded that the maximum height difference of jacking and the speed of jacking are in the first two places in the ranking of each graph, and the difference in values with other features is large, which can be concluded that the maximum height difference of jacking and the speed of jacking are the most important feature variables affecting the safety performance (stress) of the structure, which is also in line with the previous research on what should be paid attention to during the construction process, and also in line with the construction specification mentioned in the content to be controlled. Therefore, in projects where the overall jacking method is used, extra attention should be paid to the performance of both, and the difference between the maximum and minimum heights of the jacking frame should be strictly controlled, and the jacking should be carried out slowly and at an even speed.

In addition, compared with the maximum difference in height and speed of jacking, the maximum jacking height and average height have less influence on the structural safety compared with the maximum difference in height and speed of jacking, which also proves that as long as the difference in height of the jacking frame and the speed of jacking can be controlled, the safety performance and structural condition can be guaranteed.

4. Security control decision-making objectives

4.1. Minimum stress change target

In engineering projects, construction safety has always been a problem that has aroused widespread concern, and so far there is still no widely recognised structural safety risk indicator system, but some experts and scholars have also put forward safety risk indicators that they believe have some value, this paper is based on the construction simulation of the stress change of the key rods as one of the optimisation objectives, aiming to reasonably control the height and speed of the jacking frame, to The purpose of this paper is to keep the stress changes generated during the jacking process of the structure within the permissible range and reduce the stress peaks and fluctuations as much as possible, so as to protect the structure from safety problems and ensure the quality and safety of the construction, as shown in Eq. 4 or Eq. 5.

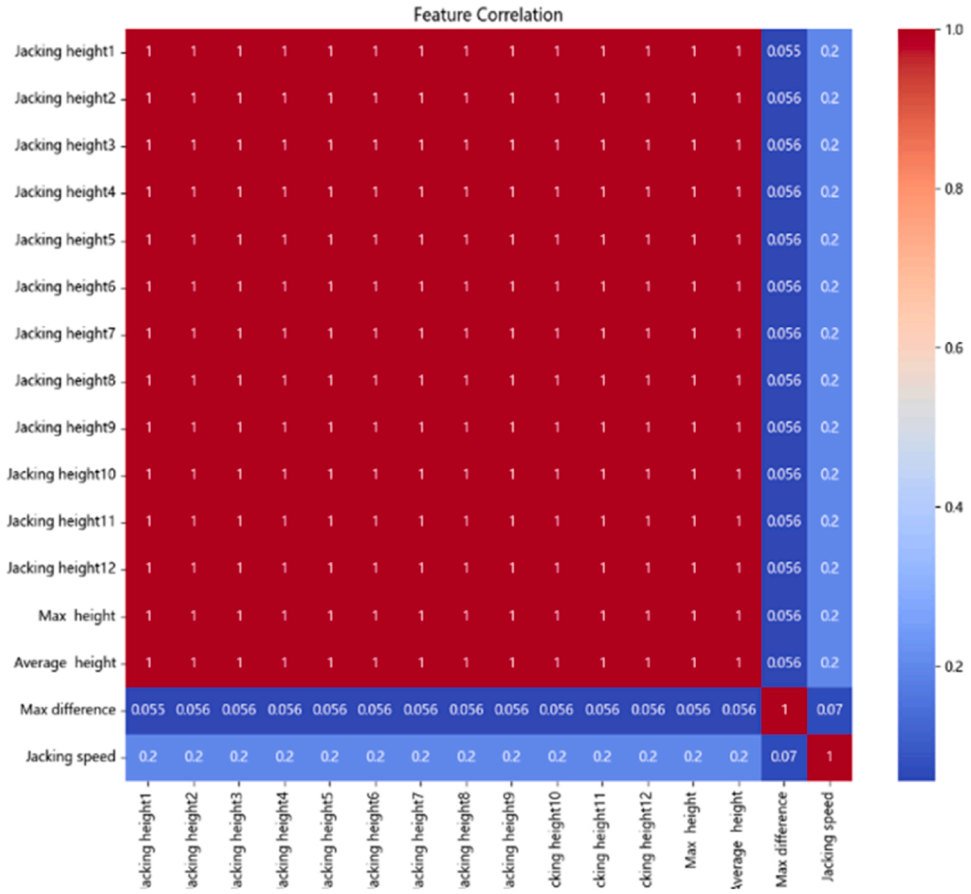


Fig. 6. Feature Correlation.

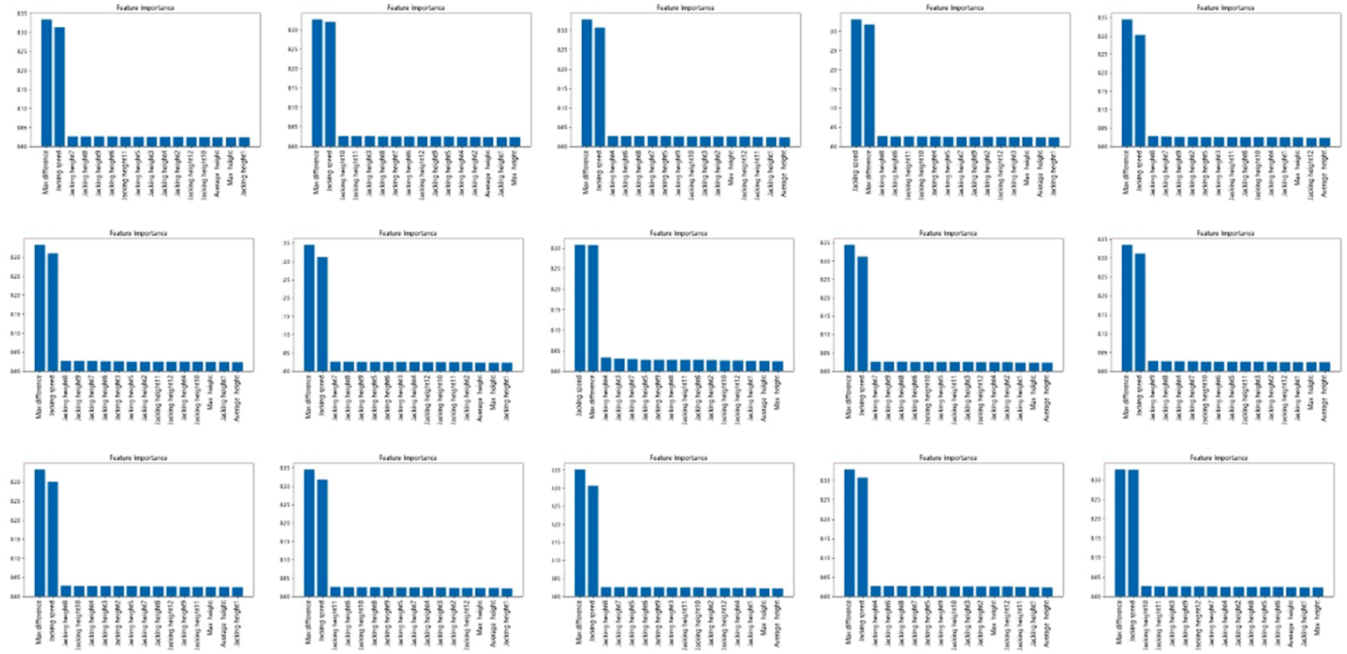


Fig. 7. Feature lmportance.

$$\sigma_{\min} = Min \sum_{i=1}^{n} |\Delta\sigma_i| \tag{4}$$

Or

$$\sigma_{\min} = Min\left(\max_i |\Delta\sigma_i|\right) \tag{5}$$

$\Delta\sigma_i$ is the stress change in bar i during jacking, and the minimum stress change objective can be expressed as minimising the total or peak stress change, depending on the needs of the project.

4.2. Minimum jacking frame height difference target

In the construction process of overall jacking method, due to the hydraulic jacking device's oil output is mechanical and part of the jacking position characteristics, it is not possible to ensure that all jacking frame uniform force, and also can not ensure that the height of the steel roof rise synchronously without any difference. And due to the influence of the difference in the value of the lift, it will cause the structural rods and jacks to be unevenly stressed and cause harm. The main factor affecting the deflection of the mesh frame is caused by the asynchrony during jacking, followed by the stiffness of the columns. It is more important to control the jacking height difference in order to reduce the offset of the net frame. Therefore, reasonable control of the height difference of the jacking frames aims to ensure that the height of each jacking frame is kept within the error range during the jacking process, so as to avoid the structural deformation, stress concentration and other construction safety problems caused by the excessive height difference, as shown in Eq. 6.

$$\Delta h_{j,k} < \max_{j,k} h_{j,k} - \min_{j,k} h_{j,k} < \Delta h_{\max} \tag{6}$$

Among them, $\Delta h_{j,k}$ is the height difference between the highest point and the lowest point of the jacking frame, and the maximum allowable value of the height difference is $\Delta h_{\max}$, j,k is the number of different jacking frames, and $h_{j,k}$ is the height of the jacking frame corresponding to the number of the jacking frame, so the allowable difference of each jacking point is 1/1000 of the spacing between adjacent jacking pillars, and not more than 10 mm.

When the net frame is jacked up as a whole, the permissible difference of each jacking point is 1/1000 of the spacing of each adjacent jacking pillar, and is not more than 10 mm, so it is necessary to limit the height difference to control the jacking synchronicity, in the process of jacking up the net frame of the project, when the height difference between any two points in the 12 points reaches 5 mm, the control system puts forward a command on the hydraulic pumping station, which makes the high position solenoid valve shut down automatically, and then the jacking low point and the jacking low point will be closed automatically. When the difference in height between the low point and the high point is zero, the solenoid valve at the high point will be reconnected to continue the jacking of the point.

4.3. Overall jacking speed target

Under the premise of ensuring structural safety and construction efficiency, through reasonable control of the jacking speed and the degree of coordination between the jacking frames, the steel structure rises smoothly and slowly during the jacking process, minimising the vibration and dynamic response of the structure during the jacking process, preventing excessive structural acceleration due to the overall jacking of the jacking frames too fast, or structural instability due to a single overly large jacking height, and controlling the overall jacking speed. The purpose is to make the stress distribution of the structure more uniform during the jacking process, avoid structural damage caused by local stress concentration, and ensure the smoothness and controllability of the jacking process.

4.4. Control system

The overall jacking method in this paper adopts multiple hydraulic jacking frames to provide jacking force in groups, and according to the stress collected by the sensors and the height linkage of the jacking frames, the steel mesh frames are assembled and shaped in-situ, and then jacked up to the unloading height by the jacking system, so as to realise the overall jacking of the large-span steel mesh frames structure.

The overall synchronous jacking system consists of jacking system (jacking frame group), power system (hydraulic pump station), monitoring system (sensors) and control system (computer) composed of four parts, the overall jacking synchronous control flowchart as shown in Fig. 8.

The jacking frame group is the jacking component of the system, which can be configured in different quantities and arranged in the corresponding positions according to the load of the steel mesh roof required by the project, and the counterforce at each jacking point can be accurately obtained through the finite element analysis of the digital twin physical model, which is used in the hydraulic jacking jacks in this project.

The hydraulic pump station provides power for the whole jacking system, and the jacking and descending of the jacking frame through the extension cylinder action or contraction cylinder action is divided into total control and sub-control, which carries out the overall synchronous jacking control and the sub-control of each jacking frame locally, which can effectively improve the synchronous adjustment performance of the whole jacking system.

The monitoring system obtains mechanical information (stress, displacement, etc.) by deploying sensors around each jacking frame, obtains displacement information through the computer control system, and provides real-time feedback of mechanical and displacement data to the digital twin model. The digital twin model regulates the next action according to the current structural air attitude information.

The main control computer sets the stress limit value according to the result of finite element construction process analysis. When the stress near the jacking point is too large or the height difference between individual jacking frame and other jacking frames is too large, it will automatically alarm and send out the shutdown command, and adjust the height of individual jacking frames so that the whole structure is at the same height again.

5. Conclusions and outlook

In this paper, the digital twin-driven safety control method is applied to safety monitoring and intelligent control by taking the construction of the overall jacking of the large-span steel roof cover in the process of the reconstruction and expansion of an airport as an example. By constructing a digital twin model, the stress, strain and other parameters monitored in real time are analysed and fault diagnosis is performed.



Fig. 8. Nodal diagram of bolt ball connection between jacking frame and top chord of net frame.

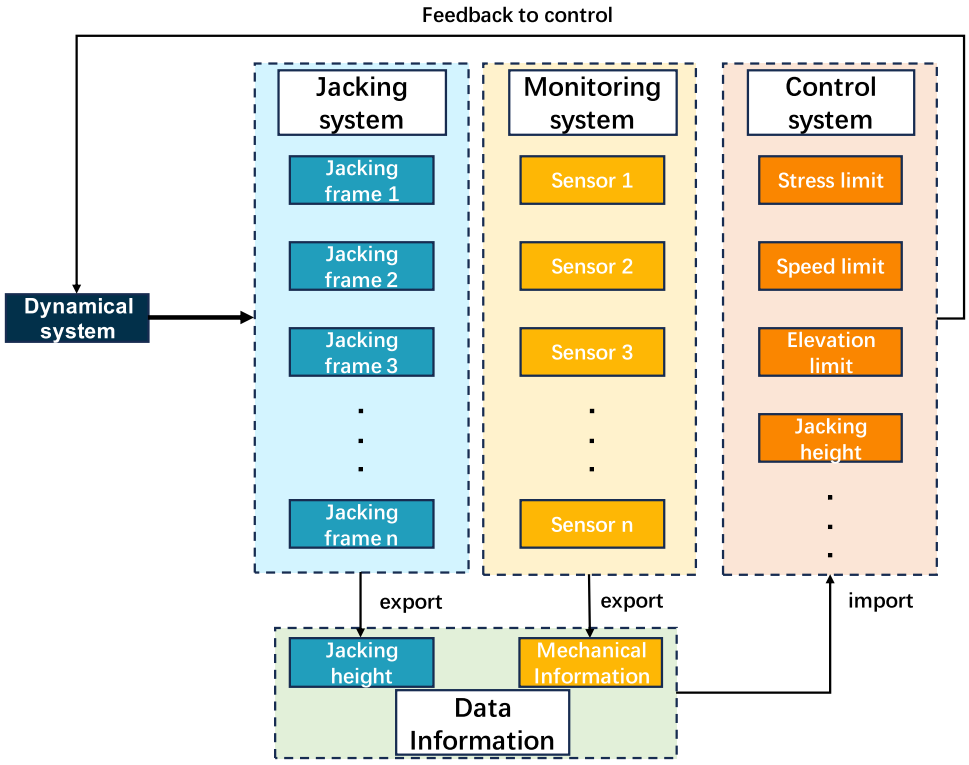


Fig. 9. Integral jacking synchronization control flow chart.

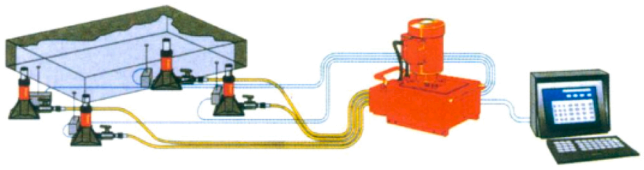


Fig. 10. Integral jacking synchronization control system connection diagram.

After practical application verification, the method can effectively improve the safety and reliability of the large-span steel structure, reduce the maintenance cost and extend the service life. The specific conclusions are as follows:

(1) The digital twin-driven safety and intelligent control method for large-span steel structures is an effective solution that can achieve real-time monitoring, data analysis, fault diagnosis and predictive maintenance of large-span steel structures. With the continuous development and improvement of digital twin technology, its application in the field of large-span steel structures will become more and more extensive. In the future, the in-depth integration of digital twin technology with advanced information technologies such as Internet of Things, cloud computing and big data can be further explored to provide more comprehensive, efficient and intelligent support for the safe and intelligent control of the construction process of large-span steel structures.

(2) Through the data collected in the actual engineering scenarios and data analysis based on the random forest algorithm, the height difference of jacking frame and jacking speed are derived from the characteristic correlation analysis as the biggest factors affecting the mechanical performance of the structure, which is consistent with the content of the specification, proving the feasibility of the method and verifying its applicability in the construction safety. Therefore, through real-time monitoring and predictive control, the constructor can observe more intuitively the dynamic change of the overall jacking behaviour of the steel mesh frame during the construction stage, which significantly strengthens the structural safety in the construction process.

(3) In order to ensure the analysis accuracy of the model, the data collected during the construction process is used. The efficiency of data transmission also needs to be improved. Future research could focus on improving the quality and integration of the model data, such as optimising predictive control algorithms, using high-precision sensors and data high-rate transmission techniques to obtain more comprehensive construction data. In addition, the method can be extended to different types of structures and other construction scenarios using monolithic jacking or monolithic jacking, such as bridges with high-rise building platforms.

CRediT authorship contribution statement

Zhansheng Liu: Writing – review & editing, Project administration, Investigation, Funding acquisition. **Chengkuan Ji:** Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Zhang Qingwen:** Formal analysis, Visualization, Writing – review & editing.

Declaration of Competing Interest

The author declares no conflict of interest.

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References

[1] Zhou G, Song Q, Li A, Shen S, Zhou Q, Wang B. Assessment on the progressive collapse resistance of a long-span curved spatial grid structure with main trusses. Ksce J Civ Eng 2021;26:1239–53. https://doi.org/10.1007/s12205-021-5371-1.

[2] Zhang A, Wang X, Liu X, Shu W, Zhu Z, Qiang S, et al. Shaking table test on overall-scale model of long-span steel structure of Beijing Daxing International Airport terminal. J Build Struct 2021;42:1–13. https://doi.org/10.14006/j.jzjgxb.2020.C088 (in Chinese).

[3] Zhao Y, Zhang Y, Xu J, Yu P, Sun B. Shaking table test on seismic performance of integrated station-bridge high-speed railway station. Structures 2022;46:1981–93. https://doi.org/10.1016/j.istruc.2022.11.042.
[4] Li J, Qu Z, Wang T. Lessons from the seismic behavior of a steel grid roof structure heavily damaged in Lushan earthquake. Earthq Eng Eng Vib 2019;18:95–111. https://doi.org/10.1007/s11803-019-0492-y.
[5] Wang W, Nie G, Bai W, Xu W, Du K. Study on the strong earthquake failure mechanism of space grid structure considering rotational ground motion. Structures 2023;57:105140. https://doi.org/10.1016/j.istruc.2023.105140.
[6] Teng J, Lu W, Cui Y, Zhang R. Temperature and displacement monitoring to steel roof construction of shenzhen bay stadium. Int J Struct Stab Dyn 2016;16:1640020. https://doi.org/10.1142/s0219455416400204.
[7] Deng D, Zou J, Cai J, Chen Z, Wang H. Non-contact measurement method of displacement of spherical joints in long-span space steel structures. Structures 2024;61:106089. https://doi.org/10.1016/j.istruc.2024.106089.
[8] Liu Y-F, Liu X-G, Fan J-S, Spencer BF, Wei X-C, Kong S-Y, et al. Refined safety assessment of steel grid structures with crooked tubular members. Autom Constr 2019;99:249–64. https://doi.org/10.1016/j.autcon.2018.11.027.
[9] Yan S, Liu W, Song G, Zhao P, Zhang S. Connection looseness detection of steel grid structures using piezoceramic transducers. 155014771875923 Int J Distrib Sens Netw 2018;14. https://doi.org/10.1177/1550147718759234.
[10] Xu J, Xu L, Ma Q, Han Q. Force evaluation of internal cable of prestressed grids based on field monitoring and hierarchical objective model updating. J Civ Struct N a Monit 2023;13:709–27. https://doi.org/10.1007/s13349-023-00673-w.
[11] Xu C, Lei H, Wang G. Damage identification method of long-span spatial structure based on time-series model of measured data. J Civ Struct N a Monit 2023;13:693–707. https://doi.org/10.1007/s13349-023-00672-x.
[12] Wang Lei, Zhao Yuanyuan, Liu Jiaxiang. A Kriging-based decoupled non-probability reliability-based design optimization scheme for piezoelectric PID control systems. Mech Syst Signal Process 2023:110714. https://doi.org/10.1016/j.ymssp.2023.110714.
[13] Li Zeshang, Wang Lei, Gu Kaixuan. Efficient reliability-based concurrent topology optimization method under PID-driven sequential decoupling framework. Thin-Walled Struct 2024:112117. https://doi.org/10.1016/j.tws.2024.112117.
[14] Liu Jiaxiang, Wang Lei, Wu Zhangming. Novel study on ac tive reliable PID controller design based on probability density evolution method and interval-oriented sequential optimization strategy. Comput Methods Appl Mech Eng 2024:117057. https://doi.org/10.1016/j.cma.2024.117057.
[15] Li Zeshang, Wang Lei, Xinyu Geng. A level set reliability-based topology optimization (LS-RBTO) method considering sensitivity mapping and multi-source interval uncertainties. Comput Methods Appl Mech Eng 2024:116587. https://doi.org/10.1016/j.cma.2023.116587.
[16] Latsou C, Ariansyah D, Salome L, Erkoyuncu JA, Sibson J, Dunville J. A unified framework for digital twin development in manufacturing. Adv Eng Inform 2024;62:102567. https://doi.org/10.1016/j.aei.2024.102567.
[17] Li L, Lei B, Mao C. Digital twin in smart manufacturing. J Ind Inf Integr 2022;26:100289. https://doi.org/10.1016/j.jii.2021.100289.
[18] Liu Z, Liu Z, SUN J, DU X. Intelligent construction methods and model experiments based on digital twins. J Build Struct 2021;42:26–36. https://doi.org/10.14006/j.jzjgxb.2020.0599 (in Chinese).
[19] Jiang S, Song H, Huang X, Chen H, Wang W. Digital Twin Driven Intelligent Assessment Method for Structural Safety. China Civ Eng J 2024:1–14. https://doi.org/10.15951/j.tmgcxb.24040296 (in Chinese).
[20] Liu Y, Sun Y, Yang A, Gao J. Digital twin-based ecogreen building design. Complex 2021;2021:1–10. https://doi.org/10.1155/2021/1391184.
[21] Jiang Y, Li M, Guo D, Wu W, Zhong RY, Huang GQ. Digital twin-enabled smart modular integrated construction system for on-site assembly. Comput Ind 2022;136:103594. https://doi.org/10.1016/j.compind.2021.103594.
[22] Chacón R, Posada H, Ramonell C, Jungmann M, Hartmann T, Khan R, et al. Digital twinning of building construction processes. case study: a reinforced concrete cast-in structure. J Build Eng 2024;84:108522. https://doi.org/10.1016/j.jobe.2024.108522.
[23] Jiao Z, Du X, Liu Z, Liu L, Sun Z, Shi G. Sustainable operation and maintenance modeling and application of building infrastructures combined with digital twin framework. Sensors 2023;23:4182. https://doi.org/10.3390/s23094182.
[24] Xia J, Zou G. Operation and maintenance optimization of offshore wind farms based on digital twin: A review. Ocean Eng 2023;268:113322. https://doi.org/10.1016/j.oceaneng.2022.113322.
[25] Ohueri CC, Masrom MAN, Habil H, Ambashe MS. IoT-based digital twin best practices for reducing operational carbon in building retrofitting: a mixed-method approach. Eng, Constr Archit Manag 2023. https://doi.org/10.1108/ecam-08-2023-0827.
[26] Mafipour MS, Vilgertshofer S, Borrmann A. Automated geometric digital twinning of bridges from segmented point clouds by parametric prototype models. Autom Constr 2023;156:105101. https://doi.org/10.1016/j.autcon.2023.105101.
[27] Jiang F, Ma L, Broyd T, Chen K, Luo H. Underpass clearance checking in highway widening projects using digital twins. Autom Constr 2022;141:104406. https://doi.org/10.1016/j.autcon.2022.104406.
[28] Ling J, Li X, Li H, Shen Y, Rui Y, Zhu H. Data acquisition-interpretation-aggregation for dynamic design of rock tunnel support. Autom Constr 2022;143:104577. https://doi.org/10.1016/j.autcon.2022.104577.
[29] Li X, Tang L, Ling J, Chen C, Shen Y, Zhu H. Digital-twin-enabled JIT design of rock tunnel: Methodology and application. Tunn N a Technol 2023;140:105307. https://doi.org/10.1016/j.tust.2023.105307.
[30] Liu Z, Shi G, Du X, Sun Z, Jiao Z. Intelligent safety control method of prestressed steel structure driven by digital twins. J Tianjin Univ(Sci Technol) 2023;56:1043–53 (in Chinese).
[31] Wang Z, Zhang Z, Liu Z, Dezhkam M, Zhao Y. Methods and applications of safety control for cable net structure considering spatiotemporal changes. Sustainability 2023;15:13922. https://doi.org/10.3390/su151813922.
[32] Wang L, Liu H, Zhang F, Guo L, Chen Z. Spatial structure digital twins: application in intelligent health monitoring of cable dome structures. Autom Constr 2024;165:105489. https://doi.org/10.1016/j.autcon.2024.105489.
[33] Leo B. Random forests. Mach Learn 2001.
[34] Zhang Z, Luo Y. Restoring method for missing data of spatial structural stress monitoring based on correlation. Mech Syst Signal Process 2017;91:266–77. https://doi.org/10.1016/j.ymssp.2017.01.018.